

Groups, rings and modules

OLIVER WEST

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Lecture 1 – 22/01/26

§1 Groups – a reminder

Definition 1.1. A *group* is a triple $(G, \cdot, e)$ where G is a set,

$$\cdot : G \times G \rightarrow G$$

is a function, and $e \in G$ such that

- *Associativity.* $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
- *Identity.* For all $a \in G$, $a \cdot e = e \cdot a = a$;
- *Inverses.* For all $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

The size of G , $|G|$, is called the *order* of the group.

Proposition 1.2 (Uniqueness of inverses)

Inverses are unique.

Proof. Left as an exercise (see 1A). ■

Definition 1.3 (Subgroups). If G is a group and $H \subseteq G$, then we say H is a *subgroup* if

- $e \in H$;
- for all $a, b \in H$, $a \cdot b \in H$;
- $(H, \cdot, e)$ is a group.

In order to say item 3, we must first establish 1 and 2.

Proposition 1.4

A non-empty subset $H \subseteq G$ is a subgroup if and only if, for all $h_1, h_2 \in H$, $h_1 h_2^{-1} \in H$.

Proof. Left as an exercise (see 1A). ■

Definition 1.5. A group G is called *abelian* if, for all $g_1, g_2 \in G$, $g_1 g_2 = g_2 g_1$.

Example 1.6 • We have $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}/n\mathbb{Z}$.

- The symmetric groups S_n : permutations of $\{1, \dots, n\}$ under composition. More generally for any set X , we have $\text{Sym}(X)$.
- The dihedral groups D_{2n} : the group of symmetries of a regular n -gon.
- The general linear group over $\mathbb{R}$, $\text{GL}(n, \mathbb{R})$ or $\text{GL}_n(\mathbb{R})$, is the set of $\{f : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid f \text{ is an } \mathbb{R}\text{-linear bijection}\}$. Since $\text{GL}(n, \mathbb{R})$ is made of functions, we can think of it a subset of $\text{Sym}(\mathbb{R}^n)$. (Equivalently, it's the set of n by n invertible matrices with elements in $\mathbb{R}$).

Some examples of subgroups:

- The even permutations $A_n \leq S_n$ (the alternating group);
- Rotational symmetries of an n -gon, $C_n \leq D_{2n}$;
- Determinant one matrices $\text{SL}(n, \mathbb{R}) \leq \text{GL}(n, \mathbb{R})$.

Other examples include $C_2 \times C_2$, $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Definition 1.7. Let G be a group, $g \in G$ and $H \leq G$ a subgroup. Then the *left coset*

$$gH = \{gh : h \in H\}$$

Basic properties include that $g \in gH$, so cosets are non-empty. Also, cosets partition G – they have no non-trivial overlaps and every g lies in some coset. Furthermore, the map

$$\begin{aligned} H &\rightarrow gH \\ h &\mapsto gh \end{aligned}$$

defines a bijection.

Theorem 1.8 (Lagrange)

If G is a finite group and $H \leq G$ then

$$|G| = |H| \cdot |G : H|$$

where $|G : H|$, the index of H in G , is the number of cosets of H in G .

Definition 1.9. Let G be a group and let $g \in G$. The *order* of g , $\text{ord}(g)$, is the smallest positive integer n such that $g^n = e$, or infinity if no such n exists.

Proposition 1.10

Let $g \in G$. Then, $\text{ord}(g) \mid |G|$.

Proof. Let $g \in G$ and consider $H = \{e, g, g^2, \dots, g^{n-1}\}$ where $n = \text{ord}(g)$. This is a subgroup: it is non-empty, and $g^r \cdot g^{-s} = g^{r-s} \in H$ (after potentially multiplying by $g^n = e$). Also, the elements of H are distinct – if $g^i = g^j$ for $0 \leq i < j \leq n$, then $g^{j-i} = e$, contradicting the minimality of $\text{ord}(g)$. Hence $|H| = \text{ord}(g)$ – now apply Lagrange. ■

§2 Normal subgroups, quotients and homomorphisms

Let $H \leq G$. When are two cosets of H in G equal? If $gH = g'H$, then $g \in g'H$, so $g = g'h$ for some $h \in H$. In other words, $gH = g'H \iff (g')^{-1}g \in H$.

We denote the set of left cosets of $H \leq G$ by $G/H = \{gH : g \in G\}$. Does G/H carry a natural group structure?

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We propose the ‘natural’ group structure:

$$g_1H \cdot g_2H = (g_1g_2)H$$

But this rule may depend on the coset representatives we pick. We must check that it doesn't. Suppose $g_2H = g'_2H$, so $g'_2 = g_2h$ for some $h \in H$. Then

$$g_1H \cdot g'_2H = (g_1g'_2)H = g_1g_2hH = g_1g_2H$$

On the other hand, if $g_1H = g'_1H$, so $g'_1 = g_1h$ for $h \in H$. Then

$$g'_1H \cdot g_2H = g_1hg_2H$$

and that is not automatically equal to g_1g_2H – we need $g_2^{-1}hg_2 \in H$ to hold for all $g_2 \in G, h \in H$.

Definition 2.1 (Normal subgroups). A subgroup $H \leq G$ is *normal* if, for all $g \in G, h \in H, ghg^{-1} \in H$. We write this as $H \triangleleft G$.

Remark 2.2. When $H \triangleleft G$, then G/H inherits a group structure by the rule above. We will refer to this as the *quotient group* of G by H .

Definition 2.3 (Homomorphisms). Let G and H be groups. A homomorphism from G to H is a function $f : G \rightarrow H$ such that, for all $g_1, g_2 \in G$, $f(g_1g_2) = f(g_1)f(g_2)$.

Results such as $f(e_G) = e_H$ follow immediately.

Lemma 2.4

If $f : G \rightarrow H$ is a homomorphism, then $f(g^{-1}) = f(g)^{-1}$.

Proof. Calculate $f(g \cdot g^{-1})$ in two ways.

$$\textcircled{1} \quad f(gg^{-1}) = f(e_G) = e_H;$$

$$\textcircled{2} \quad f(gg^{-1}) = f(g)f(g^{-1})$$

so $f(g)f(g^{-1}) = e_H$ and the result follows. ■

Definition 2.5. Let $f : G \rightarrow H$ be a homomorphism. The *kernel* of f is

$$\ker f = \{g \in G : f(g) = e_H\}$$

and the *image* of f is

$$\text{Im } f = \{h \in H : h = f(g) \text{ for some } g\}$$

Proposition 2.6

Let $f : G \rightarrow H$ be a homomorphism. Then $\ker f \triangleleft G$ and $\text{Im } f \leq H$.

Proof. Observe $e \in \ker f$ so $\ker f$ non-empty. Now, if $x, y \in \ker f$, then

$$f(xy^{-1}) = f(x)f(y)^{-1} = e$$

so $xy^{-1} \in \ker f$. For normality, let $x \in G$ and $g \in \ker f$. Now

$$f(xgx^{-1}) = f(x)ef(x)^{-1} = e$$

so $xgx^{-1} \in \ker f$ and hence $\ker f \triangleleft G$.

For the image, observe that, if $a, b \in \text{Im } f$ with $a = f(x)$, $b = f(y)$, then $ab^{-1} = f(xy^{-1}) \in \text{Im } f$. ■

Remark. Observe $f : G \rightarrow G/H$ taking $g \mapsto gH$ is a homomorphism. This is a nice way of expressing its ‘naturalness’.

Definition 2.7. A homomorphism $f : G \rightarrow H$ is an *isomorphism* if it is a bijection. Two groups G and H are said to be *isomorphic*, $G \cong H$, if there exists an isomorphism between them.

Theorem 2.8 (First isomorphism theorem)

Let $f : G \rightarrow H$ be a homomorphism, then $\ker f$ is normal in G and the homomorphism

$$\begin{aligned} G/\ker f &\rightarrow \text{Im } f \\ g\ker f &\mapsto f(g) \end{aligned}$$

is an isomorphism. In particular, $G/\ker f \cong \text{Im } f$.

Proof. We have already shown the normality result. Consider the map

$$\begin{aligned}\phi : G/\ker f &\rightarrow \operatorname{Im} f \\ g\ker f &\mapsto f(g)\end{aligned}$$

- ϕ is well-defined. If $g\ker f = g'\ker f$, then $g'g^{-1} \in \ker f$, so $f(g'g^{-1}) = e$ which gives $f(g) = f(g')$ as required.
- ϕ is surjective. Let $h \in \operatorname{Im} f$. Then $h = f(g)$ for some $g \in G$, and so $h = \phi(g\ker f)$.
- ϕ is injective. Suppose $\phi(g\ker f) = \phi(g'\ker f)$, then $f(g) = f(g')$. But then $g'g^{-1} \in \ker f$, so $g\ker f = g'\ker f$.

■

Example 2.9

Consider the homomorphism

$$\begin{aligned}\exp : (\mathbb{C}, +) &\rightarrow (\mathbb{C}^*, \times) \\ z &\mapsto e^z\end{aligned}$$

Then $\ker \exp = 2\pi i\mathbb{Z}$, so $\mathbb{C}^* \cong \mathbb{C}/2\pi i\mathbb{Z}$.

Theorem 2.10 (Second isomorphism theorem)

Let $H \leq K$ and $K \triangleleft G$. Then

$$HK = \{hk : h \in H, k \in K\}$$

is a subgroup of G . Furthermore, $H \cap K$ is normal in H , and there is an isomorphism $HK/K \cong H/H \cap K$.

Proof. We will show the results in order.

- HK is a subgroup. It contains e_G , so is non-empty. Now consider $x = hk$ and $y = h'k'$. We'll show $yx^{-1} \in HK$. Observe

$$\begin{aligned}yx^{-1} &= h'k'(hk)^{-1} \\ &= h'k'k^{-1}h^{-1} \\ &= (h'h^{-1})h(k'k^{-1})h^{-1} \\ &= h'h^{-1}k''\end{aligned}$$

by normality of K .

■